KAIZEN MATHS

Lesson starter

Algebraic substitution

Three variables

4 questions · about 5 minutes

Name _____

Evaluate each expression. Use brackets for negative values.

S1 Given $a = 3, b = 3, c = 1$

$$3a - 6b + 2c$$

S2 Given $a = 0, b = 0, c = 0$

$$5a - 4b + 2c$$

S3 Given $a = 0, b = -1, c = -2$

$$4a - 4b + 4c$$

S4 Given $a = 3, b = -4, c = -3$

$$6a - 3b + 5c$$

Show your working

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S4 Given $a = 3, b = -4, c = -3$

$6a - 3b + 5c$

AS-04 · S · Starter

Show your working

KAIZEN MATHS

Exit ticket

Algebraic substitution

Three variables

Name _____

Evaluate each expression.

E1 Given $a = 3, b = -4, c = -3$

$$6a - 3b + 5c$$

E2 Given $a = 3, b = 0, c = -5$

$$4a - 3b + 5c$$

E3 Given $a = 1, b = 3, c = -3$

$$4a - 3b + 5c$$

AS-04 · E · 3 questions

Show your working

KAIZEN MATHS

Exit ticket

Algebraic substitution

Three variables

Name _____

Evaluate each expression.

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$6a - 3b + 5c$

E2 Given $a = 3, b = 0, c = -5$

$4a - 3b + 5c$

E3 Given $a = 1, b = 3, c = -3$

$4a - 3b + 5c$

AS-04 · E · 3 questions

Show your working

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Algebraic substitution

Three variables

Name _____

Evaluate each expression.

E1 Given $a = 3, b = -4, c = -3$

$6a - 3b + 5c$

E2 Given $a = 3, b = 0, c = -5$

$4a - 3b + 5c$

E3 Given $a = 1, b = 3, c = -3$

$4a - 3b + 5c$

AS-04 · E · 3 questions

Show your working

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Exit ticket

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Evaluate each expression.

E1 Given $a = 3, b = -4, c = -3$

$6a - 3b + 5c$

E2 Given $a = 3, b = 0, c = -5$

$4a - 3b + 5c$

E3 Given $a = 1, b = 3, c = -3$

$4a - 3b + 5c$

AS-04 · E · 3 questions

Show your working

Teacher answers & working

AS-04 · Three variables

Page 1: two starters. Page 2: four exit tickets. Cut along the dashed lines.

Print student pages single-sided at 100%. Match S and E labels to the solutions below.

S1

Given a=3, b=3, c=1

$$3a - 6b + 2c$$

Substitute values.

$$3\times 3 - 6\times 3 + 2\times 1$$

$$9 - 18 + 2$$

$$-7$$

S2

Given a=0, b=0, c=0

$$5a - 4b + 2c$$

Substitute values.

$$5\times 0 - 4\times 0 + 2\times 0$$

$$0$$

$$0$$

S3

Given a=0, b=-1, c=-2

$$4a - 4b + 4c$$

Substitute values.

$$4\times 0 - 4\times (-1) + 4\times (-2)$$

$$0 + 4 - 8$$

$$-4$$

S4 / E1

Given a=3, b=-4, c=-3

$$6a - 3b + 5c$$

Substitute values.

$$6\times 3 - 3\times (-4) + 5\times (-3)$$

$$18 + 12 - 15$$

$$15$$

E2

Given a=3, b=0, c=-5

$$4a - 3b + 5c$$

Substitute values.

$$4\times 3 - 3\times 0 + 5\times (-5)$$

$$12 - 25$$

$$-13$$

E3

Given a=1, b=3, c=-3

$$4a - 3b + 5c$$

Substitute values.

$$4\times 1 - 3\times 3 + 5\times (-3)$$

$$4 - 9 - 15$$

$$-20$$